

Manejo Intensivo da Pancreatite Aguda

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Classificação de gravidade

JAMA | Review

Acute Pancreatitis A Review

Michael A. Mederos, MD; Howard A. Reber, MD; Mark D. Girgis, MD

Leve → Sem disfunção orgânica

Moderada → Disfunção orgânica < 48h

Grave → Disfunção orgânica > 48h

UTI



Classificação e Predição de gravidade

Table 1. Modified Marshall Scoring System for Organ Dysfunction

Organ system	Score ^a				
	0	1	2	3	4
Respiratory (Pao ₂ /Fio ₂) ^b	>400	301-400	201-300	101-200	<101
Kidney (serum creatinine), µmol/L	<134	134-169	170-310	311-439	>439
Kidney (serum creatinine), mg/dL	<1.4	1.4-1.8	1.9-3.6	3.7-4.9	>4.9
Cardiovascular (systolic blood pressure), mm Hg	>90	<90, fluid responsive	<90, not fluid responsive	<90, pH <7.3	<90, pH <7.2

JAMA. 2021;325(4):382-390.
doi:10.1001/jama.2020.20317



Classificação e Predição de gravidade

APACHE II	BISAP (2008, Gut)	Ranson
• Age • Temperature • Mean arterial pressure • pH • Heart rate • Respiratory rate • Sodium • Potassium • Creatinine • Acute kidney failure • Hematocrit • WBC count • Glasgow Coma Scale • FIO₂	• BUN >25 mg/dL (>8.9 mmol/L) • Impaired mental status • >2 SIRS criteria • Age >60 y • Pleural effusion present	Admission: • Age >55 y • WBC count >16 000/μL • Lactate dehydrogenase >350 U/L • AST >250 U/L • Glucose >200 mg/dL Within 48 h: • Fall in hematocrit >10% • Increase in BUN >5 mg/dL • Calcium <8 mg/dL • Pao₂ <60 mm Hg • Base deficit >4 mEq/L • Fluid loss >6 L

Prediction of mortality ²²	Score ≥8	Score ≥3	Score ≥3
• Sensitivity (95% CI) • Specificity (95% CI)	• 0.95 (0.77-1.00) • 0.68 (0.63-0.73)	• 0.56 (4.23-7.55) • 0.91 (0.90-0.91)	• 0.93 (0.78-0.99) • 0.69 (0.65-0.79)

JAMA. 2021;325(4):382-390.
doi:10.1001/jama.2020.20317



Pilares do manejo intensivo :

Analgesia

Hidratação

Nutrição

Vigilância de complicações locais



Analgesia

Dipirona + Paracetamol

AINE

Opióides

Estratégias poupadoras de
opióides

Narcotic Analgesic Effects on the Sphincter of Oddi: A Review of the Data and Therapeutic Implications in Treating Pancreatitis

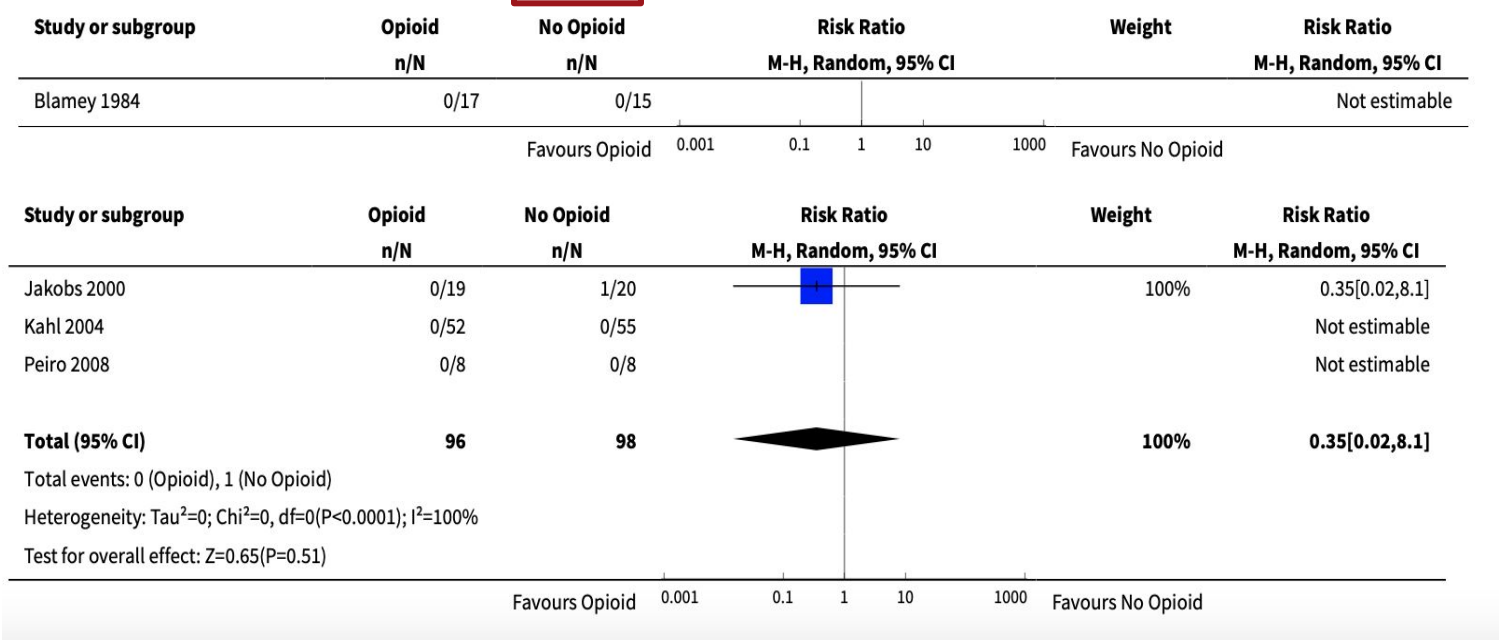
Donald R. Thompson, M.D.

*Gastroenterology Division, Department of Internal Medicine, Baylor University Medical Center,
Dallas, Texas*



Analgesia - Opióides

Analysis 4.4. Comparison 4 Opioids versus no opioids, Outcome 4 Number of deaths from any cause (secondary outcome 3).



Opioids for acute pancreatitis pain (Review)

Basurto Ona X, Rigau Comas D, Urrútia G



Analgesia

Dipirona + Paracetamol

Opióides



Estratégias poupadoras de
opióides

Cetamina EV
Lidocaína EV
Gabaérgicos
Bloqueio epidural
GRUPO DA DOR !



Analgesia

1. Individualizar sempre;
2. Acessar intensidade da dor (escalas) e gravidade da pancreatite;
3. Ponderar usar analgésicos mais fortes (step down) para alívio imediato da dor;
4. Estratégias poupadoras de opioide.

Hidratação

The NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

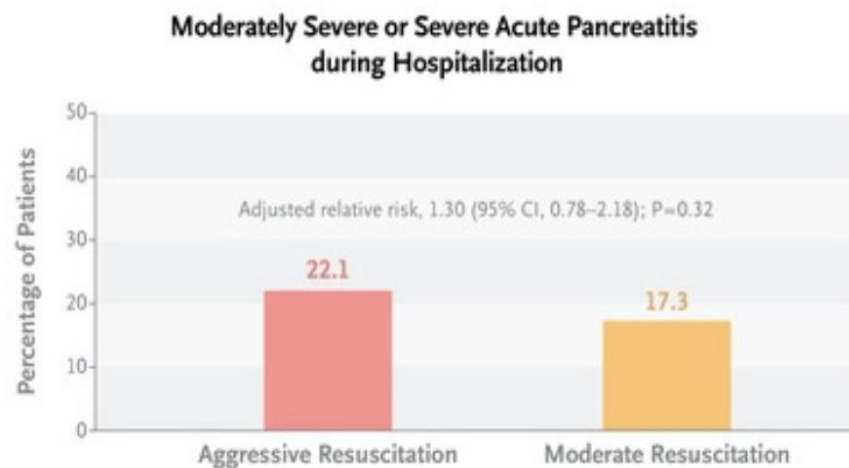
Aggressive or Moderate Fluid Resuscitation in Acute Pancreatitis

E. de-Madaria, J.L. Buxbaum, P. Maisonneuve, A. García García de Paredes,

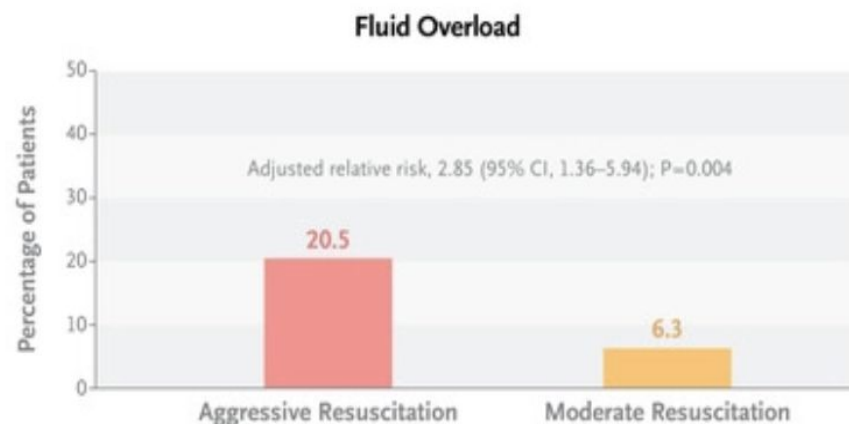
Agressivo: Bolus 20ml/kg > manter 3ml/kg 12h
Ex. 70kg >> Bolus 1400ml e manter 2520 em 12h
~ 6L em 24h

Moderado: metade
Ex.: 70kg >> Bolus 750ml, manter 1260 em 12h
~3L em 24h





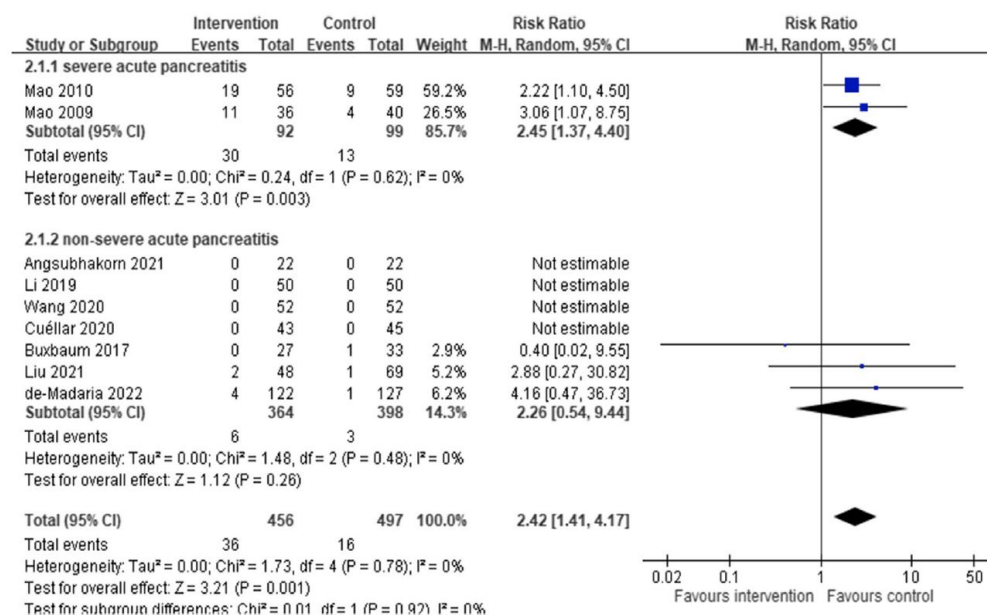
“...the use of aggressive fluid resuscitation led to a higher risk of volume overload and did not show the hypothesized benefit in disease-specific outcomes.”





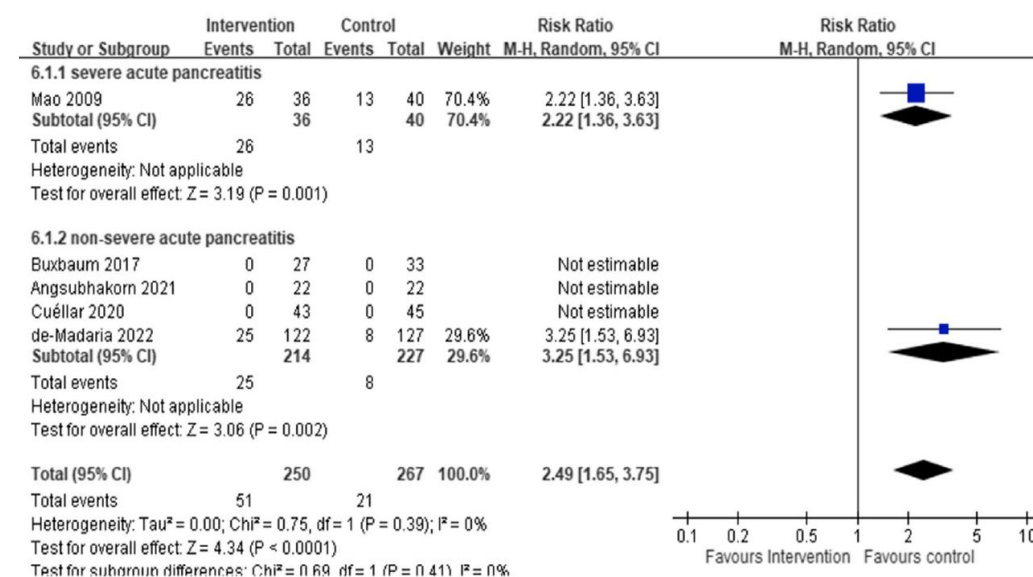
Comparison of clinical outcomes between aggressive and non-aggressive intravenous hydration for acute pancreatitis: a systematic review and meta-analysis

Mortalidade

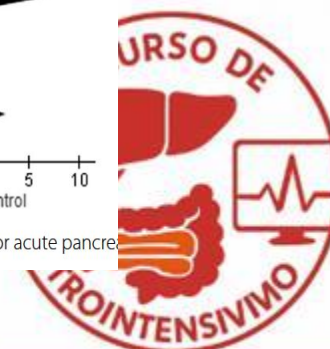


Mortality risk, comparing aggressive (intervention) and non-aggressive (control) hydration protocols for acute pancreatitis

Congestão



Fluid-related complication risk, comparing aggressive (intervention) and non-aggressive (control) hydration protocols for acute pancreatitis



RANDOMIZED CONTROLLED TRIAL

Balanced Solution versus Normal Saline in Predicted Severe Acute Pancreatitis

A Stepped Wedge Cluster Randomized Trial

Annals of Surgery ():10.1097/SLA.0000000000006319, May 6, 2024. | DOI: 10.1097/SLA.0000000000006319

endpoints, patients who received BMES had less systemic inflammatory response syndrome(19/112,17.0% versus 43/147,29.3%, $P=0.024$) and increased organ failure-free days (3.9 d(SD2.7) versus 3.5days(SD2.7), $P<0.001$ by trial day7. They also spent more time alive and out of ICU(26.4 d(SD5.2) versus 25.0days(SD6.4), $P=0.009$) and hospital(19.8 d(SD6.1) versus16.3days(SD7.2), $P<0.001$) by trial day30.



AGA SECTION

American Gastroenterological Association Institute Guideline on Initial Management of Acute Pancreatitis



Seth D. Crockett,¹ Sachin Wani,² Timothy B. Gardner,³ Yngve Falck-Ytter,^{4,5} and Alan N. Barkun⁶;
on behalf of American Gastroenterological Association Institute Clinical Guidelines Committee



Recommendation	Strength of recommendation	Quality of evidence
1A. In patients with AP, the AGA suggests using goal-directed therapy for fluid management. <i>Comment: The AGA makes no recommendation whether normal saline or Ringer's lactate is used.</i>	Conditional	Very low



Goal-directed therapy ??

PAM

DIURESE

Ht

FC



VVS

Fluid
challenge

POCUS



Nutrição - Qual via ?

Leve

Via oral o quanto antes

Moderada / Grave

?

Historicamente, muita NPT...



Nutrição - Qual via ?

A Randomized Controlled Trial of Enteral versus Parenteral Feeding in Patients with Predicted Severe Acute Pancreatitis Shows a Significant Reduction in Mortality and in Infected Pancreatic Complications with Total Enteral Nutrition

Maxim S. Petrov Mikhail V. Kukosh Nikolay V. Emelyanov

Department of Surgery, Nizhny Novgorod State Medical Academy, Nizhny Novgorod, Russia

NPT >
mortalidade

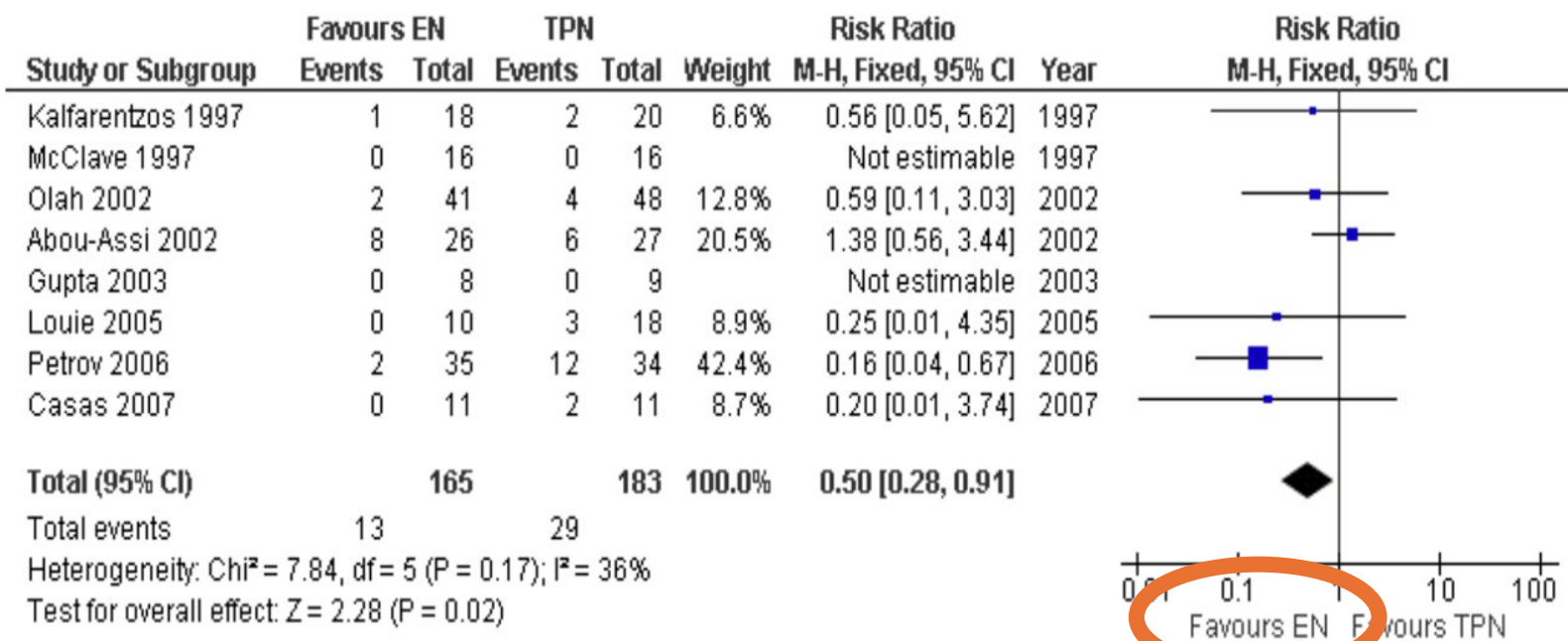
infected pancreatic necrosis. The death rate was significantly ($p = 0.003$) greater among the patients that received TPN (35%, 12/34) while for the patients that received TEN it was only 6% (2 of 35).

Dig Surg 2006;23:336-3.



Nutrição - Qual via ?

Figure 3. Forest plot of comparison: 1 Enteral versus parenteral nutrition for acute pancreatitis, outcome: 1.1 Mortality.



NPT com + Infecções ?

Table 2. Number of infectious complications of severe AP in the two groups of patients

Complications	TEN (n = 35)	TPN (n = 34)	p
<i>Pancreatic infection</i>	7	16	0.02
Infected pancreatic necrosis	5	12	NA
Pancreatic abscess	2	4	NA
<i>Extrapancreatic infection</i>	4	11	0.04
Urinary infection	2	4	NA
Pneumonia	2	2	NA
Infections of the central venous catheter	–	5	NA



Nutrição - Quando ?

ORIGINAL ARTICLE

Early versus On-Demand Nasoenteric Tube Feeding in Acute Pancreatitis

Table 2. Primary and Secondary End Points, According to the Intention-to-Treat Analysis.*

Outcome	Early Tube Feeding (N=101)	On-Demand Tube Feeding (N=104)	Risk Ratio (95% CI)	P Value
Primary composite end point: infection or death — no. (%)	30 (30)	28 (27)	1.07 (0.79–1.44)	0.76
Secondary end points				
Infection — no. (%)†	25 (25)	27 (26)	0.97 (0.70–1.34)	0.87
Infected pancreatic necrosis	9 (9)	15 (14)	0.74 (0.43–1.26)	0.28
Bacteremia	17 (17)	18 (17)	0.98 (0.68–1.43)	1.00
Pneumonia	12 (12)	13 (12)	0.97 (0.63–1.50)	1.00
Death — no. (%)	11 (11)	7 (7)	1.27 (0.85–1.89)	0.33
Necrotizing pancreatitis — no. (%)‡	64 (63)	65 (62)	1.06 (0.77–1.47)	0.76

Dieta SNE em 24h

VS.

Dieta VO por
aceitação até 72h



Nutrição - Pós pilórica ?

Early Enteral Nutrition in Severe Acute Pancreatitis: A Prospective Randomized Controlled Trial Comparing Nasojejunal and Nasogastric Routes

*Ajay Kumar, DM, Namrata Singh, MSc, Shyam Prakash, PhD, Anoop Saraya, DM,
and Yogendra Kumar Joshi, MD, PhD*

TABLE 2. Comparison of Study Parameters (Primary Endpoints)
in the Two Groups

	NJ (Group 1) (n = 14)	NG (Group 2) (n = 16)	P Value
Recurrence of pain	1	1	1.000
Partial parenteral nutrition	4	6	0.709
Hospital stay (days)*	29.93 ± 25.54	24.06 ± 14.35	0.437
Infection rate			
Blood culture positive	2	3	0.467
Tracheal aspirate	1	0	
Pancreatic aspirate	3	3	
Bile culture	0	1	
Outcome			
Discharge	10	11	0.761
Surgery	2	1	
Death	4	5	



ESPEN Guideline

ESPEN guideline on clinical nutrition in acute and chronic pancreatitis

Marianna Arvanitakis ^{a,*}, Johann Ockenga ^b, Mihailo Bezmarevic ^c, Luca Gianotti ^d,
Željko Krznarić ^e, Dileep N. Lobo ^{f,g}, Christian Löser ^h, Christian Madl ⁱ, Remy Meier ^j,
Mary Phillips ^k, Henrik Højgaard Rasmussen ^l, Jeanin E. Van Hooft ^m, Stephan C. Bischoff ⁿ

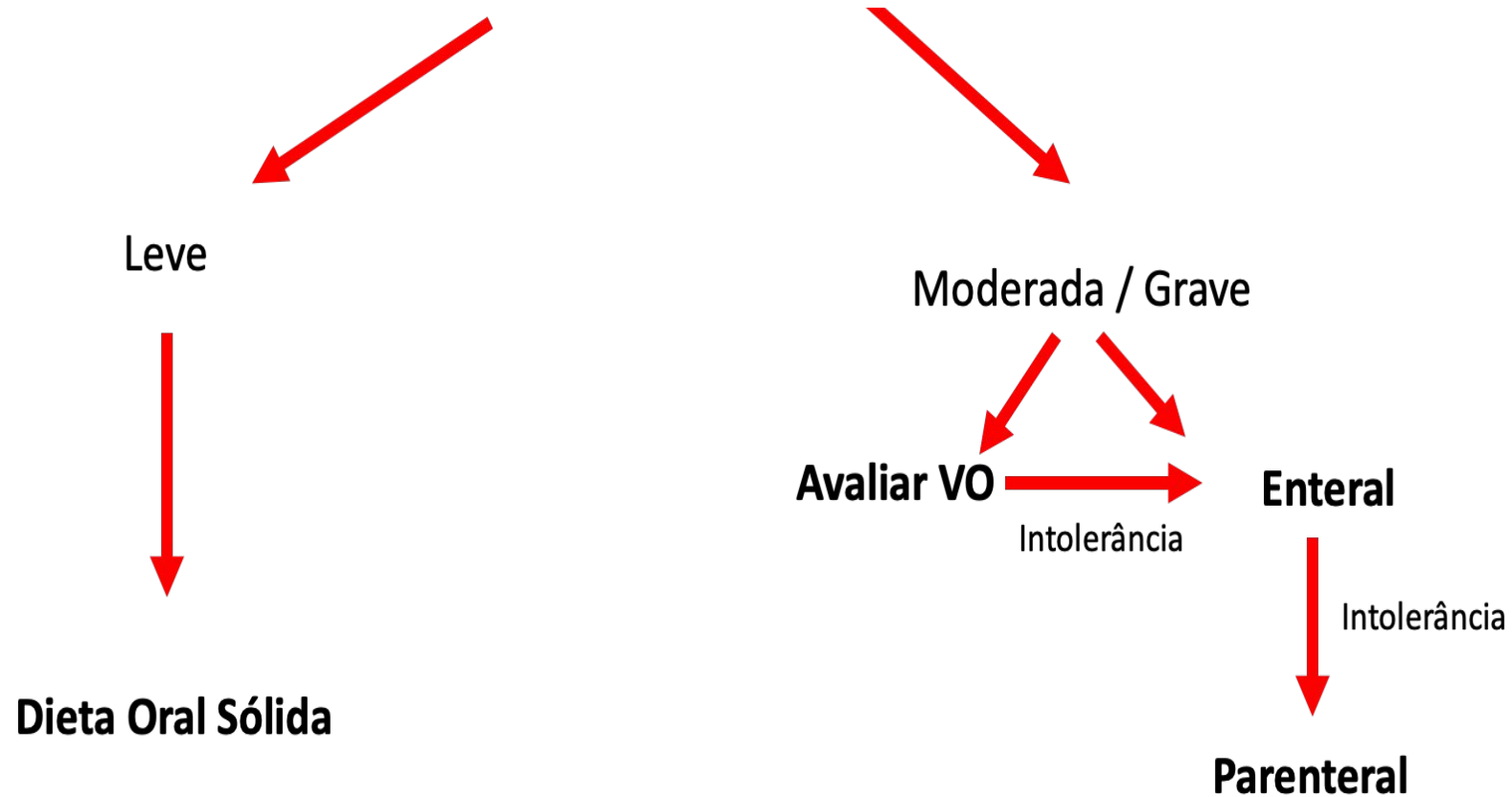


If EN is required in patients with AP, it should be administered via a nasogastric tube. Administration via a nasojejunal tube should be preferred in case of digestive intolerance.

Grade of Recommendation B – Strong consensus (95% agreement).



Pancreatite Aguda



Antibióticos

Early Antibiotic Treatment for Severe Acute Necrotizing Pancreatitis

A Randomized, Double-Blind, Placebo-Controlled Study

Annals of Surgery • Volume 245, Number 5, May 2007

“Patients with 30% necrosis of the pancreas confirmed by contrast-enhanced computerized tomography (CT) were eligible...”



Antibióticos

Early Antibiotic Treatment for Severe Acute Necrotizing Pancreatitis

A Randomized, Double-Blind, Placebo-Controlled Study

Annals of Surgery • Volume 245, Number 5, May 2007

TABLE 2. Development and Time to Onset of Pancreatic or Peripancreatic Infection From Symptom Onset

	Treatment Group			
	Meropenem (n = 50)		Placebo (n = 50)	
	n	%	n	%
Patients with pancreatic or peripancreatic infection	9	18	6	12
Patients with resistant pancreatic or peripancreatic infection	4	8	3	6
Mean (range) no. of days to diagnosis of infection	21.3 (5–35)	—	20.8 (11–25)	



Antibióticos

Comparison 1. Antibiotics versus control

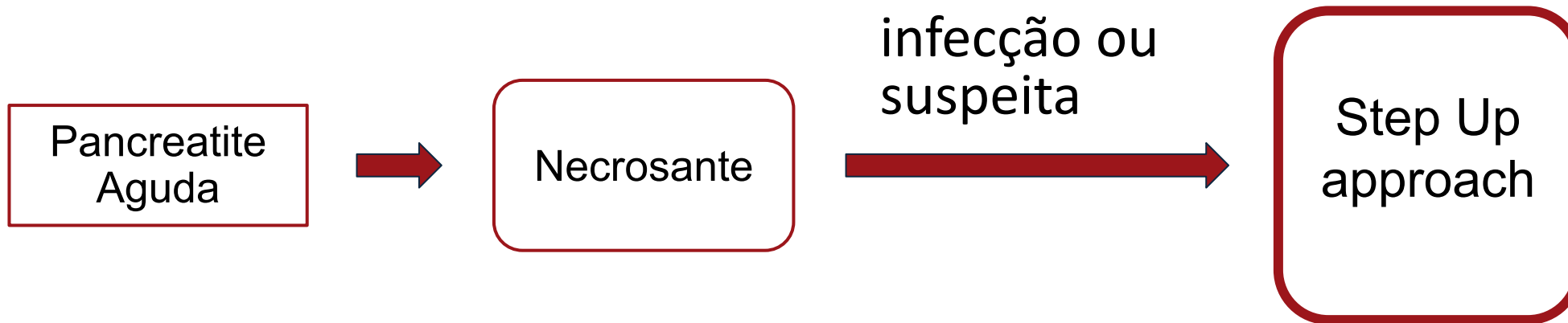
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1 Mortality	7	404	Risk Ratio (M-H, Fixed, 95% CI)	0.60 [0.34, 1.05]
2 Infected Pancreatic Necrosis	7	404	Risk Ratio (M-H, Random, 95% CI)	0.85 [0.57, 1.26]
3 Non-Pancreatic Infections	5	318	Risk Ratio (M-H, Random, 95% CI)	0.62 [0.36, 1.06]
4 All sites infections	5	318	Risk Ratio (M-H, Random, 95% CI)	0.69 [0.44, 1.09]
5 Fungal Infection	7	404	Risk Ratio (M-H, Random, 95% CI)	1.06 [0.41, 2.70]
6 Operative Treatment	6	378	Risk Ratio (M-H, Random, 95% CI)	0.90 [0.62, 1.31]

Antibiotic therapy for prophylaxis against infection of pancreatic necrosis in acute pancreatitis (Review) Copyright © 2010 The Cochrane Collaboration. Published by John Wiley & Sons, Ltd.



Complicações locais e intervenção

Sempre o mais conservador possível !



ORIGINAL ARTICLE

A Step-up Approach or Open Necrosectomy for Necrotizing Pancreatitis

Hjalmar C. van Santvoort, M.D., Marc G. Besselink, M.D., Ph.D.,

“The step-up approach consisted of percutaneous drainage followed, if necessary, by minimally invasive retroperitoneal necrosectomy”.

Table 3. Primary and Secondary End Points.*

Outcome	Minimally Invasive Step-up Approach (N=43)	Primary Open Necrosectomy (N=45)	Risk Ratio (95% CI)	P Value
Primary composite end point: major complications or death — no. (%)†	17 (40)	31 (69)	0.57 (0.38–0.87)	0.006
Secondary end points				
Major complication — no. (%)				
New-onset multiple-organ failure or systemic complications‡	5 (12)	19 (42)	0.28 (0.11–0.67)	0.001
Multiple-organ failure	5 (12)	18 (40)		



Immediate versus Postponed Intervention for Infected Necrotizing Pancreatitis

L. Boxhoorn, S.M. van Dijk, J. van Grinsven, R.C. Verdonk, M.A. Boermeester,

October 7, 2021

N Engl J Med 2021; 385:1372-1381

DOI: 10.1056/NEJMoa2100826

Chinese Translation 中文翻译

Necrose Infectada

Drenar Imediato

Aguardar organizar



Immediate versus Postponed Intervention for Infected Necrotizing Pancreatitis

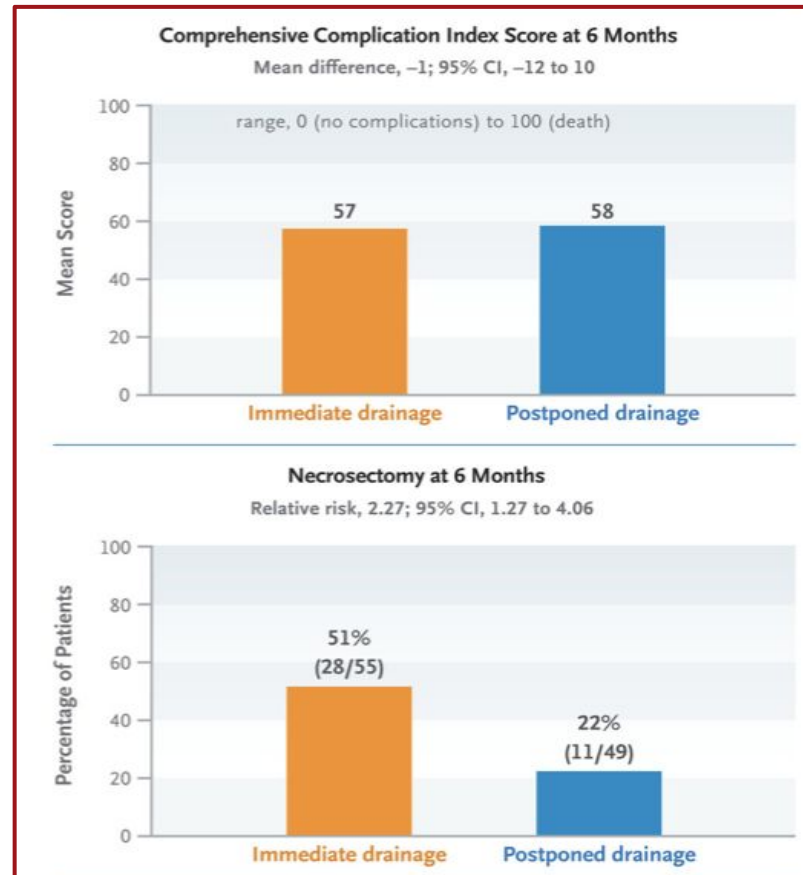
L. Boxhoorn, S.M. van Dijk, J. van Grinsven, R.C. Verdonk, M.A. Boermeester,

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Chinese Translation 中文翻译



Early versus delayed catheter drainage for patients with necrotizing pancreatitis and early persistent organ failure (TIMING): a multicenter randomized controlled trial

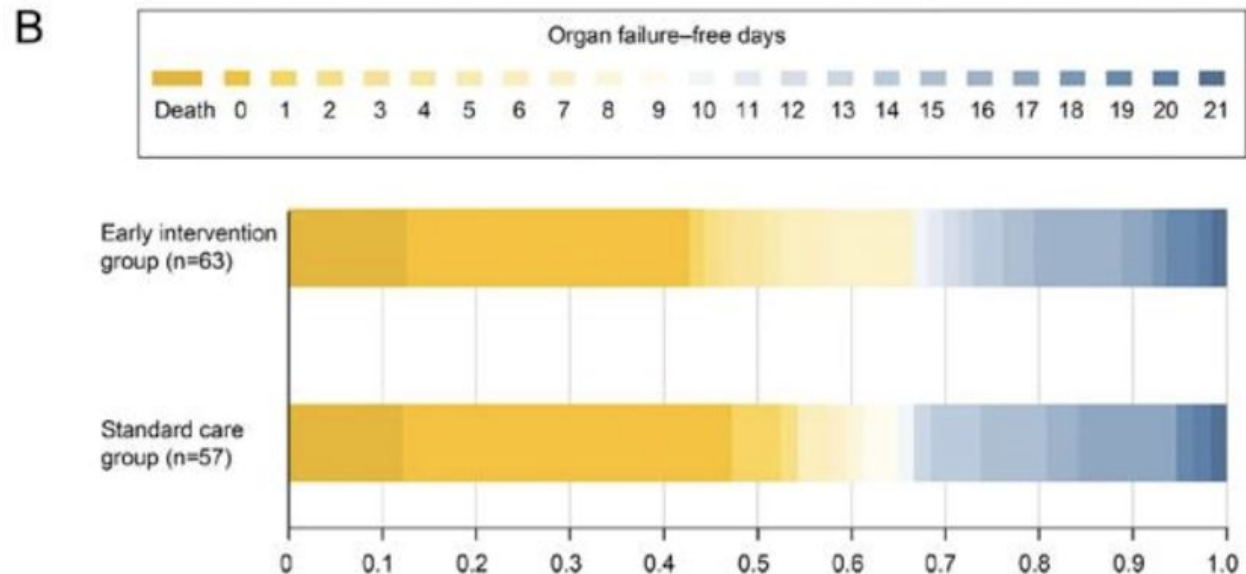
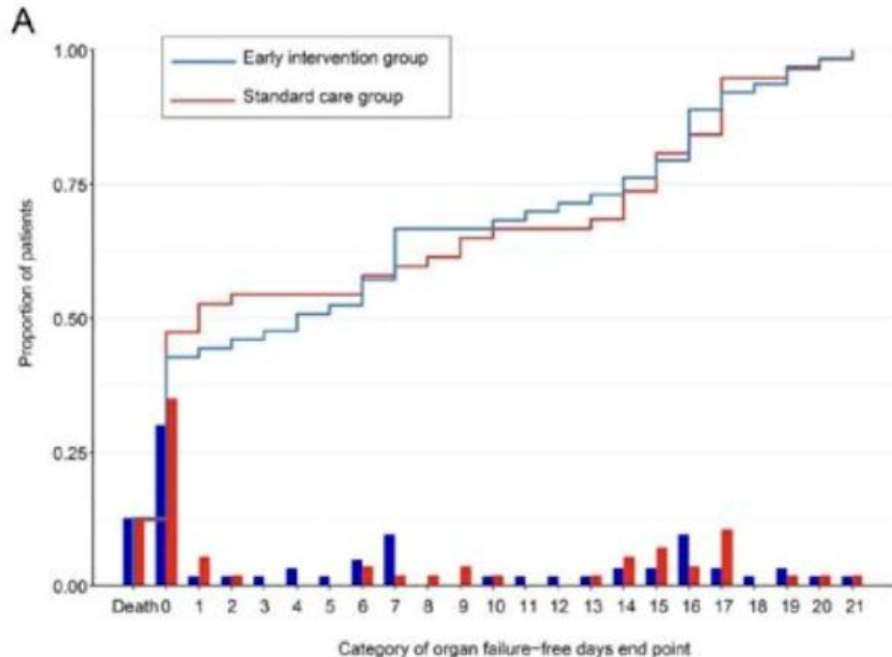
Original | Published: 14 July 2025

(2025) [Cite this article](#)



Intensive Care Medicine

ANC no 7 dia + disfunção persistente >>> Drenar imediato ou aguardar ??



Conclusão

Analgesia

- Individualizar
- Opióides podem ser usados

Hidratação

- Evitar hiper hidratação
- Guiar por metas : clínicas + laboratoriais + novas tecnologias

Nutrição

- Via oral / enteral de preferência
- Conforme tolerância

ATB

- Só quando infecção documentada
- Carbapenêmicos ainda são a escolha



Conclusão

Colecistectomia

- Antes da alta
- *Etiologia biliar

Coleção infectada

- Drenagem minimamente invasiva
- Sem pressa / deixar organizar ?



Obrigada

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